Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/21/2005
PI Name:	Poirier, David Sc.D.		
Project Title:	Comparison of Structure and Segregation in Alloys Directionally Solidified in Terrestrial and Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85721-0012	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	06/01/1996	**End Date:**	06/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/01/2005 in June 2006 per PI (original end date was 6/01/2006)--jp 2/2009		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Heinrich, J. (University of Arizona) Tewari, S. (Cleveland State University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Terrestrial solidification experiments on/ Pb-Sb alloys have been done using different growth conditions to effect different dendritic microstructures. Numerical modeling to simulate the transport phenomena and solidification of the planned experiments is currently underway at both MSFC and the University of Arizona (UA). Terrestrial solidification experiments are underway at Cleveland State University (CSU). As ingot-samples are prepared at CSU, they will be characterized for dendrite morphology, macrosegregation, and microsegregation. The researchers at UA are also simulating dendritic growth in the presence of convection. This is a program to conduct experiments in a long duration microgravity environment, with the objective of studying dendritic microstructures and segregation in directionally solidified dendritic alloys. The research builds on our previous grant in which we generated an extensive data-base on directionally solidified Pb-Sn alloys and simulated solidification of the alloys using continuum theory of porous media. Based on our previous research, it is clear that thermosolutal convection masks the effects of diffusional transport during solidification and also leads to macro-segregates known as freckles. We are convinced, therefore, that with terrestrial experiments, diffusive and convective phenomena occur simultaneously, which complicate the study of dendritic growth. In microgravity, however, it would be possible to effect directional solidification with no thermosolutal convection. Our hypotheses are the convection and attendant macrosegregation will be eliminated and the dendritic microstructure will be greatly altered in microgravity. Models of dendritic solidification of alloys rely on the assumption of diffusional transports of both heat and solute. However, the thermosolutal convection of liquid in, from, or close to the solid plus liquid region of solidifying alloys masks the diffusional processes at the dendrite tips and is responsible for most macrosegregation in castings and ingots. Availability of microgravity provides an opportunity to obtain experimental data, where thermosolutal convection is negligible. With this grant, growth conditions for such microgravity experiments will be defined by combining mathematical modeling and simulations with the results of terrestrial experiments. Thermosolutal convection will be significantly reduced in microgravity, and thus we can compare segregation and microstructure in specimens terrestrially solidified and those solidified in microgravity. In terrestrial experiments, dendritic solidification is difficult to study because specimens are plagued with severe forms of macro segregation caused by thermosolutal convection, including: 1) continuous segregation from end-to-end; 2) occurrence of localized segregates referred to as freckles; and, 3) dendrite clustering. In the microgravity environment, the thermosolutal convection will be greatly diminished, and the convection will be confined mainly to the minor flows of interdendritic liquid required to satisfy solidification shrinkage and possibly due to "g-jitter." Morphological features, metrics, and segregation in samples directionally solidified in an earth-based furnace and in samples grown in microgravity are expected to be quite different. The technological significance of the research is that it will provide scientific underpinning to our understanding and control of how single-crystal castings solidify. Single-crystal castings are critical components in high-temperature gas-turbine engines that are used in high-speed aircraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The researchers at UA simulated dendritic growth at the microstructural level in the presence of convection for possible application in understanding welding in low-gravity and/or extra-terrestrial environments. We have begun to set our simulator for dendritic growth into three dimensions, made possible by effecting a collaboration with an expert in finite-element meshing at Centro Atomico in Argentina. In earlier work measurements of dendrite specific surface area (Sv) in the quenched mushy-zone of directionally solidified samples were used to relate the specific surface area of the dendritic solid to the primary dendrite arm spacing and the volume fraction of the interdendritic liquid. Incorporation of this relationship into the Kozney-Carman permeability of a mushy zone yields a relationship that can be used to define a mushy-zone Rayleigh number. During the past year longitudinal macrosegregation and channel-segregate data available were examined to see whether the defined Rayleigh number can be used to quantitatively explain macrosegregation defects. It was observed that this Rayleigh number does show a strong correlation between increasing interdendritic thermosolutal convection and the extent of experimentally observed longitudinal macrosegregation in a range of directionally solidified hypoeutectic Pb-Sb and Pb-Sn alloys grown at varying thermal gradients and growth speeds. This Rayleigh number is, therefore, useful in identifying the processing parameters that are likely to produce channel segregates during directional solidification.
Bibliography Type:	Description: (Last Updated: 06/06/2025)
Articles in Peer-reviewed Journals	Tewari SN, Tiwari R, Magadi G. "Mushy-zone Rayleigh number to describe macrosegregation and channel segregate formation during directional solidification of metallic alloys." Metallurgical and Materials Transactions A, 2004 Sep;35(9):2927-34. http://dx.doi.org/10.1007/s11661-004-0240-3 , Sep-2004
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Papers from Meeting Proceedings	Westra DG, Heinrich JC, Poirier DR. "Simulating the Effect of Space Vehicle Environments on Directional Solidification of a Binary Alloy." 42nd AIAA Aerospace Sciences Meeting, Reno, Nevada, Jan. 5-8, 2004. 42nd AIAA Aerospace Sciences Meeting, Reno, Nevada, Jan. 5-8, 2004. AIAA paper 2004-630. , Jan-2004